

● EASY

● ALGEBRA

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# Systems of two linear equations in two variables

# 04

6 questions · ~9 min

Q1

ALGEBRA

Connor has  $c$  dollars and Maria has  $m$  dollars. Connor has 4 times as many dollars as Maria, and together they have a total of \$ 25.00. Which system of equations represents this situation?

- A.  $c = 4m$   
 $c + m = 25$
- B.  $m = 4c$   
 $c + m = 25$
- C.  $c = 25m$   
 $c + m = 4$
- D.  $m = 25c$   
 $c + m = 4$

A dance teacher ordered outfits for students for a dance recital. Outfits for boys cost \$26, and outfits for girls cost \$35. The dance teacher ordered a total of 28 outfits and spent \$881. If  $b$  represents the number of outfits the dance teacher ordered for boys and  $g$  represents the number of outfits the dance teacher ordered for girls, which of the following systems of equations can be solved to find  $b$  and  $g$  ?

A.  $26b + 35g = 28$   
 $b + g = 881$

B.  $26b + 35g = 881$   
 $b + g = 28$

C.  $26g + 35b = 28$   
 $b + g = 881$

D.  $26g + 35b = 881$   
 $b + g = 28$

$$y = 4$$

$$x = y + 6$$

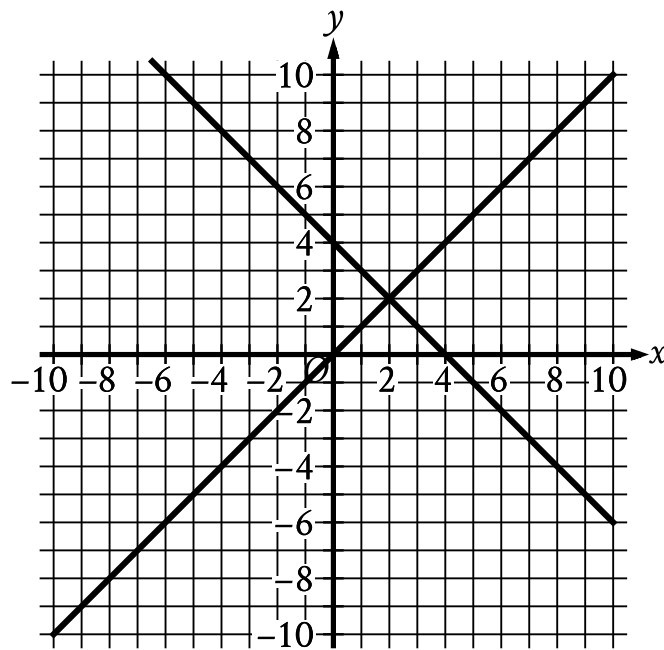
The solution to the given system of equations is  $x, y$ . What is the value of  $x$ ?

A. 10

B. 6

C. 4

D. 2



- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (0 comma 4)
    - (2 comma 2)
    - (4 comma 0)
- For the second line in the system:
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (0 comma 0)
    - (2 comma 2)
    - (4 comma 4)

The graph of a system of two linear equations is shown. What is the solution  $x, y$  to the system?

A. 0,4

B. 2, 2

C. 4, 0

D. 4, 4

Q5

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$$x = 5$$

$$y = x - 8$$

Which of the following points  $x, y$  is the solution to the given system of equations in the  $xy$ -plane?

A. 0, 0

B. 5, -3

C. 5, -8

D. 5, 8

$$3x = 12$$

$$-3x + y = -6$$

The solution to the given system of equations is  $x, y$ . What is the value of  $y$ ?

- A. -3
- B. 6
- C. 18
- D. 30